

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

GATEWAY MODULE (GWM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed via the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

The table below lists all diagnostic trouble codes (DTCs) that could be logged in the gateway module, for additional diagnosis and testing information refer to the relevant diagnosis and testing section.

For additional information, refer to: [Battery - SDV6 3.0L Diesel - Hybrid Electric Vehicle](#) (414-01 Battery, Mounting and Cables, Diagnosis and Testing) /

[Battery](#) (414-01 Battery, Mounting and Cables, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B11DB-49	Battery Monitoring Module "A" - Internal electronic failure	Battery monitoring system control module - Internal failure	NOTE: Disconnect the primary battery for 60 seconds before continuing Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new battery monitoring system control module as required
B11DB-87	Battery Monitoring Module "A" - Missing message	LIN 1 bus circuit fault	

NOTE:

If DTC B11DB-87 is flagged but a "charge system fault" warning / light is not present on the instrument cluster, then the likelihood is that the battery monitoring system control module is NOT the faulty part. If this DTC is logged, follow the diagnostic steps detailed in the Battery Diagnosis and Testing Section (Section 414-01) BEFORE replacing the battery monitoring system control module

- Check the integrity of the battery monitoring system control module connector and rectify as required. Refer to the electrical circuit diagrams and check the power supply circuit between the battery positive terminal and the battery monitoring system control module for short circuit to ground, open circuit, high resistance. Repair circuit as required. Clear DTC and retest

			▪ If fault persists, refer to the electrical circuit diagrams and check the LIN 1 bus circuit between the gateway module and the battery monitoring system control module for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear DTC and retest ▪ If the fault persists, install a new battery monitoring system control module as required
B12CD-16	Voltage Quality Module - Circuit voltage below threshold	▪ Voltage quality module power or ground circuit open circuit, high resistance ▪ LIN 2 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Voltage quality module internal failure	▪ Refer to the electrical circuit diagrams and check the integrity of the voltage quality module circuit connections at both ends. Rectify as required ▪ Refer to the electrical circuit diagrams and check the voltage quality module power and ground circuits for open circuit, high resistance. Repair circuit as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check the LIN 2 bus circuit between the gateway module and the voltage quality module for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear DTC and retest

			▪ If the fault persists, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault still persists, install a new voltage quality module
B12CD-19	Voltage Quality Module - Circuit current above threshold	▪ Voltage quality module power or ground circuit open circuit, high resistance ▪ LIN 2 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Voltage quality module internal failure	▪ Refer to the electrical circuit diagrams and check the integrity of the voltage quality module circuit connections at both ends. Rectify as required ▪ Refer to the electrical circuit diagrams and check the voltage quality module power and ground circuits for open circuit, high resistance. Repair circuit as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check the LIN 2 bus circuit between the gateway module and the voltage quality module for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear DTC and retest ▪ If the fault persists, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault still persists, install a new voltage quality module
B12CD-87	Voltage Quality Module - Missing	▪ Gateway module has not received expected	▪ Check the integrity of the voltage quality

	message	signals from the voltage quality module ▪ LIN 2 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Voltage quality module internal failure	module circuit connectors. Rectify as required ▪ Refer to the electrical circuit diagrams and check the LIN 2 bus circuit between the gateway module and the voltage quality module for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault persists, install a new voltage quality module
B12CD-92	Voltage Quality Module - Performance or incorrect operation	▪ Gateway module has not received expected signals from the voltage quality module ▪ LIN 2 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Voltage quality module internal failure	▪ Refer to the electrical circuit diagrams and check the integrity of the voltage quality module circuit connections at both ends. Rectify as required ▪ Check the integrity of the voltage quality module circuit connectors. Rectify as required ▪ Refer to the electrical circuit diagrams and check the LIN 2 bus circuit between the gateway module and the voltage quality module for short circuit to ground, short circuit to power,

			open circuit, high resistance. Repair circuit as required Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault persists, install a new voltage quality module
B12CD-96	Voltage Quality Module - Component internal failure	- Voltage quality module power or ground circuit open circuit, high resistance - LIN 2 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance - Voltage quality module internal failure	- Refer to the electrical circuit diagrams and check the integrity of the voltage quality module circuit connections at both ends. Rectify as required - Refer to the electrical circuit diagrams and check the voltage quality module power and ground circuits for open circuit, high resistance. Repair circuit as required - Refer to the electrical circuit diagrams and check the LIN 2 bus circuit between the gateway module and the voltage quality module for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, clear

			the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault persists, install a new voltage quality module
B13C5-92	Secondary Battery - Performance or incorrect operation	▪ Secondary battery connectors insecure ▪ Power supply distribution box circuit open circuit, high resistance ▪ Secondary battery system fault ▪ Charging system fault	▪ Check the integrity of the secondary battery circuit connections at the battery terminals and at the power supply distribution box. Rectify as required ▪ Refer to the electrical circuit diagrams and check the power supply distribution box circuit between the gateway module and the power supply distribution box for open circuit, high resistance. Repair circuit as required. Clear DTC and retest ▪ Refer to the relevant section of the workshop manual and test the secondary battery with the manufacturer approved battery health testing tool. Charge the secondary battery if necessary and retest. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new power supply distribution box as required
B1404-13	Charging System - Circuit open	▪ Battery/charging system fault	▪ Using the Jaguar Land Rover Approved Diagnostic

			Equipment, perform routine - Power Supply Service Mode Diagnostic. Clear the DTCs and retest ▪ If the fault persists, check and install a new generator as required
B1412-09	Quiescent Relay Box - Component failures	▪ Quiescent current control module power or ground circuit(s) open circuit, high resistance ▪ LIN 1 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Quiescent current control module internal failure	▪ Refer to the electrical circuit diagrams and check the quiescent current control module power and ground circuits for open circuit, high resistance. Repair circuit as required ▪ If fault persists, refer to the electrical circuit diagrams and check the LIN 1 bus circuit between the gateway module and the quiescent current control module for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new quiescent current control module
B1412-87	Quiescent Relay Box - Missing message	▪ LIN 1 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Quiescent current control module internal failure	▪ If fault persists, refer to the electrical circuit diagrams and check the LIN 1 bus circuit between the gateway module and the quiescent current

			control module for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear DTC and retest ▪ If the fault persists, install a new quiescent current control module as required
B1412-96	Quiescent Relay Box - Component internal failure	▪ Quiescent current control module - Internal failure	NOTE: If DTC B1412-96 is logged DO NOT install a new Quiescent Relay Box or GateWay module as this will NOT rectify the fault. ▪ No action necessary, ignore DTC
B1479-08	Power Supply Distribution Box - Bus signal/message failures	▪ LIN 2 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Power supply distribution box internal failure	▪ Refer to the electrical circuit diagrams and check the LIN 2 bus circuit between the gateway module and the power supply distribution box for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs, lock the vehicle and wait 2 minutes.

			Unlock the vehicle and retest. If the fault persists, install a new power supply distribution box as required
B1479-14	Power Supply Distribution Box - Circuit short to ground or open	- Power supply distribution box circuit short circuit to ground, open circuit, high resistance - Power supply distribution box internal failure	- Refer to the electrical circuit diagrams and check the power supply distribution box circuit between the gateway module and the power supply distribution box for short circuit to ground, open circuit, high resistance. Repair circuit as required - Using the manufacturer approved diagnostic system, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault still persists, install a new power supply distribution box as required
B1479-72	Power Supply Distribution Box - Actuator stuck open	- Power supply distribution box circuit short circuit to ground, open circuit, high resistance - Power supply distribution box internal failure	

			NOTE: Using the manufacturers approved diagnostic system, confirm the gateway module software installed is the latest level. Install the latest gateway module software as required. Should the DTC reoccur, follow the actions as described below. - Refer to the electrical circuit diagrams and check the power supply distribution box circuit between the gateway module and the power supply distribution box for short circuit to ground, open circuit, high resistance. Repair circuit as required - Using the manufacturer approved diagnostic system, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault still persists, install a new power supply distribution box as required
B1479-73	Power Supply Distribution Box - Actuator stuck closed	Power supply distribution box circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the power supply distribution

		Power supply distribution box internal failure	box circuit between the gateway module and the power supply distribution box for short circuit to ground, open circuit, high resistance. Repair circuit as required Using the manufacturer approved diagnostic system, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault still persists, install a new power supply distribution box as required
B1479-96	Power Supply Distribution Box - Component internal failure	- Power supply distribution box circuit short circuit to ground, open circuit, high resistance - Power supply distribution box internal failure	- Refer to the electrical circuit diagrams and check the power supply distribution box circuit between the gateway module and the power supply distribution box for short circuit to ground, open circuit, high resistance. Repair circuit as required - Using the manufacturer approved diagnostic system, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault still persists, install a new power supply distribution box as required
B1479-98	Power Supply Distribution Box - Component or	Power supply distribution box - Over temperature event	Refer to the electrical circuit diagrams and check the power

	system over temperature		supply distribution box circuit between the gateway module and the power supply distribution box for short circuit to ground, open circuit, high resistance. Repair circuit as required ▪ Allow the vehicle to cool. Using the manufacturer approved diagnostic system, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and run the engine for 5 minutes with all electrical loads on. If the fault persists, install a new power supply distribution box as required
P0620-00	Generator Control Circuit - No sub type information	▪ Primary drive belt incorrect tension ▪ LIN 3 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Generator internal failure	▪ Check the primary drive belt tension and rectify as required ▪ Refer to the electrical circuit diagrams and check the LIN 3 bus circuit between the gateway module and the generator for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform routine - Power Supply Service Mode Diagnostic. Clear the DTCs and retest ▪ If the fault persists, check and install a

			new generator as required
P065B-16	Generator Control Circuit Range/Performance - Circuit voltage below threshold	- Primary battery discharged - LIN 3 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: If P0620-00, P065C-01, P0A1A-87 or P0A3B-68 is/are also set, perform the relevant corrective action(s) first - Refer to the relevant section of the workshop manual and test the primary battery - Refer to the electrical circuit diagrams and check the LIN 3 bus circuit between the gateway module and the generator for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear DTC and retest
P065B-17	Generator Control Circuit Range/Performance - Circuit voltage above threshold	- Primary battery connectors insecure - Generator connectors insecure - LIN 3 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	

			NOTES: ▪ If P0620-00, P065C-01, P0A1A-87 or P0A3B-68 is/are also set, perform the relevant corrective action(s) first ▪ This DTC may be set during jump starting and/or battery starting ▪ Check the integrity of the primary battery connectors and rectify as required ▪ Check the integrity of the generator connectors and rectify as required ▪ Refer to the electrical circuit diagrams and check the LIN 3 bus circuit between the gateway module and the generator for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear DTC and retest
P065C-01	Generator Mechanical Performance - General electrical failure	▪ Primary drive belt incorrect tension	▪ Check the primary drive belt tension and rectify as required
P0A0F-71	Engine Failed To Start - Actuator stuck	▪ Engine system fault	▪ Using the manufacturer approved diagnostic system, check the engine control

			module for related DTCs and refer to the relevant DTC index
P0A1A-87	Generator Control Module - Missing message	▪ Primary battery connectors insecure ▪ Generator connectors insecure ▪ LIN 3 bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Generator internal failure	▪ Check the integrity of the primary battery connectors and rectify as required ▪ Check the integrity of the generator connectors and rectify as required ▪ Refer to the electrical circuit diagrams and check the LIN 3 bus circuit between the gateway module and the generator for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear DTC and retest ▪ If the fault persists, install a new generator as required. Clear DTC and retest
P0A3B-68	Generator Over Temperature - Event information	▪ Generator over temperature event	▪ Check generator wiring and connectors for evidence of heat damage. Investigate the cause of generator over temperature event. Using the manufacturer approved diagnostic system, clear the DTCs and retest
U0028-81	Vehicle Communication Bus A - Invalid serial data received	▪ Invalid data received from another control module via the high speed CAN bus (powertrain)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid

			data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0028-87	Vehicle Communication Bus A - Missing message	Missing message from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0028-88	Vehicle Communication Bus A - Bus off	NOTE: Circuit reference - PT_CAN_HI_GWM / PT_CAN_LO_GWM High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0037-81	Vehicle Communication Bus B - Invalid serial data received	Invalid data received from another control module via the high speed CAN bus (chassis)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0037-	Vehicle	Missing message from	Using the

87	Communication Bus B - Missing message	another control module via the high speed CAN bus (chassis)	manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0037- 88	Vehicle Communication Bus B - Bus off	NOTE: Circuit reference - CH_CAN_HI_GWM / CH_CAN_LO_GWM High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0046- 87	Vehicle Communication Bus C - Missing message	Missing message from another control module via the high speed CAN bus (hybrid)	NOTE: This DTC only applies to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index

U0046-88	Vehicle Communication Bus C - Bus off	High speed CAN bus (hybrid) circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: This DTC only applies to hybrid electric vehicles Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (hybrid) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0055-87	Vehicle Communication Bus D - Missing message	Missing message from another control module via the medium speed CAN bus (body)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0055-88	Vehicle Communication Bus D - Bus off	NOTE: Circuit reference - BO_CAN_HI_GWM / BO_CAN_LO_GWM Medium speed CAN bus (body) circuit short circuit to ground, short circuit to power, open circuit, high	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (body) circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		resistance	
U0064-87	Vehicle Communication Bus E - Missing message	Missing message from another control module via the medium speed CAN bus (comfort)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0064-88	Vehicle Communication Bus E - Bus off	NOTE: Circuit reference - MS CO_CAN_HI_GWM / MS CO_CAN_LO_GWM Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-46	Internal Control Module Software Incompatibility - Calibration/parameter memory failure	Gateway module is not configured correctly	NOTE: On L405 16MY the gateway module is integrated within the body control module/gateway module assembly Using the manufacturer approved diagnostic system, reconfigure

			the gateway module with the latest level software. Clear the DTCs and retest. If the fault persists, clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault persists, install a new body control module
U1A14-00	CAN Initialisation Failure - No sub type information	▪ Gateway module power or ground circuit open circuit, high resistance ▪ High speed CAN bus (chassis or powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Medium speed CAN bus (body or comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Gateway module internal failure	NOTE: On L405 16MY the gateway module is integrated within the body control module/gateway module assembly ▪ Refer to the electrical circuit diagrams and check the gateway module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis and powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic

			system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (body and comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new body control module
U2100-00	Initial Configuration Not Complete - No sub type information	■ Body control module is not configured correctly ■ Gateway module is not configured correctly	NOTE: On L405 16MY the gateway module is integrated within the body control module/gateway module assembly ■ Using the manufacturer approved diagnostic system, reconfigure the body control module with the latest level software ■ Using the manufacturer approved diagnostic system, reconfigure the gateway module with the latest level software
U2300-	Central Configuration	■ Car configuration file	■ Using the

41	- General checksum failure	mismatch with vehicle specification	manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2300-51	Central Configuration - Not programmed	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2300-54	Central Configuration - Missing calibration	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2300-56	Central Configuration - Invalid/incomplete configuration	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2300-57	Central Configuration - Invalid/incomplete software component	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2300-64	Central Configuration - Signal plausibility failure	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U3000-	Control Module -	Gateway module internal	

04	System internal failures	failure	NOTE: On L405 16MY the gateway module is integrated within the body control module/gateway module assembly Using the manufacturer approved diagnostic system, reconfigure the gateway module with the latest level software. Clear the DTCs, lock the vehicle and wait 2 minutes. Unlock the vehicle and retest. If the fault persists, install a new body control module
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